MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)

MG1DSCECT100- EMERGING ELECTRONICS

Duration: 2 hours Maximum Marks: 35

Practical Examination – Model Question Paper

Instructions:

- **Demonstrate** the given task
- **Viva session** (Minimum 5 questions from the Practical Module)
- **Lab Report**

Evaluation Criteria:

- **Viva**: 10 marks
- **Lab Report**: 10 marks
- **Demonstration**: 15 marks

Experiments: *(Choose any one)*

1. Draw a circuit diagram to verify Ohm's law.
2. Generate a sinusoidal wave having frequency 2 KHz and amplitude 10 V.
3. Study and plot the forward characteristics of PN junction Diode. Also find the static resistance, dynamic resistance and knee voltage.
4. Study and plot the characteristics of Zener Diode. Also find the static resistance, dynamic resistance and breakdown voltage.
5. Study and plot the forward characteristics of LED. Also find the static resistance and dynamic resistance.
6. Construct a Centre tapped full wave rectifier and draw its output.
7. Construct a half wave rectifier and draw its output.
8. Design and setup positive and negative clipper and plot its output waveforms.
9. Design and setup positive and negative clamper and plot the output waveforms.
10. Make an extension box with three plug sockets and three switches.
11. Design and develop a staircase lamp using two-way switches.
12. Write a simple Arduino code to blink an LED for one second.
13. Design an Arduino-based LED circuit that flashes three times on power-up.
14. Develop a program for an LED chasing effect.
15. Write an Arduino program to count pulses using an LDR.